Endogenizing Nature in Production Theory: A Framework of Economic Commensalism

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Abstract

The relationship between economic activity and the natural environment has long been marked by a persistent tension between productive expansion and ecological preservation.

Problem

Keywords: economic commensalism; environmental economics; rational incentives; commons; carbon markets.

1. Introduction

The relationship between economic activity and the natural environment has long been marked by a persistent tension between productive expansion and ecological preservation. The environment is essential to human welfare and economic production, yet the depletion of natural resources and the degradation of ecological systems can undermine the very productive capacity that economic activity seeks to expand. Conversely, production, investment, and technological progress generate employment, income, and improvements in living standards, so environmental policy cannot be evaluated independently of its effects on private incentives and productive viability. The relevant economic problem is therefore not whether production or preservation should prevail, but under what conditions the use of nature can remain compatible with the ecological functions on which present and future production depend.

Economic theory does not ignore nature. Land, natural resources, environmental costs, externalities, and common-pool resources have been extensively examined across different strands of the literature, such as Pigou (2010), Coase (1960), Hardin (1968), Ostrom (1990), and Dasgupta (2001), as we will see in detail in Section 3.

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In canonical representations of production, such as the Cobb–Douglas function, output is usually modeled as a function of capital and labor, sometimes with land or natural resources appearing as inputs, but ecological systems are not generally represented as a factor or cofactor that affects output, future productivity, strategic incentives, and intertemporal welfare. Instead, the environment often appears as a stock, a constraint, a source of externalities, or an object of corrective regulation, and this representation has economic consequences. This modeling choice is not neutral: by representing the environment primarily as a stock, constraint, or externality, standard frameworks limit the extent to which ecological functionality can enter private decision problems as a state that co-determines productivity.

When ecological functionality remains outside the private decision problem, preservation tends to appear as a legal obligation, a compliance cost, or an opportunity cost imposed on production. Its benefits are frequently delayed, uncertain, spatially diffuse, or collectively appropriated, whereas the returns from resource conversion may be immediate, measurable, and privately captured. Even an agent who recognizes the long-term importance of ecological integrity may therefore rationally choose short-term expansion when the expected private return from degradation exceeds the expected private return from preservation. The divergence is intensified by credit constraints, competitive pressures, incomplete information, weak enforcement, and uncertainty regarding future policies.

Governmental command-and-control regulation and penalties, carbon markets, environmental certification, payments for ecosystem services, and fiscal incentives translate parts of these insights into policy. These approaches have generated important improvements in specific dimensions, but they address different components of the problem and do not automatically establish a unified relationship among ecological conditions, production, private payoffs, and territorially grounded governance.

Nevertheless, the remaining gap is neither the absence of nature from economic thought nor the absence of legal, financial, institutional, or normative instruments intended to protect it. Rather, what remains insufficiently specified is an integrated set of conditions under which ecological functionality is jointly determined with production, its productive and ecosystem-service effects enter private payoffs, and governance is matched to the spatial scale of the relevant ecological processes. The gap addressed in this article is therefore one of analytical integration rather than the absence of environmental considerations from economics.

Against this background, the article seeks to resolve the following **problem**: under what conditions can ecological functionality be represented as an endogenous state of production and institutionally governed so that its maintenance becomes incentive-

compatible with private productive viability? This question replaces a simple opposition between use and preservation with a problem of institutional and productive design. It also shifts the analysis from whether economic activity causes any environmental impact to whether it systematically erodes the ecological conditions required for future production and social welfare.

The **general objective** of the article is to develop a theoretical-analytical framework that represents ecological functionality as an endogenous state of production and derives the conditions under which territorially aligned governance and private incentives can make its maintenance compatible with productive viability. On this basis, the article defines and characterizes economic commensalism as a distinct ecological-economic configuration.

More specifically, the **specific goals** are: (i) to distinguish the use of natural resources as conventional inputs from the role of ecological functionality as a state that affects production and responds to productive and preservation decisions; (ii) to examine how the spatial heterogeneity and imperfect substitutability of ecosystem services affect the appropriate scale of environmental governance and compensation; (iii) to derive the payoff conditions under which maintaining ecological functionality becomes individually rational for productive agents; (iv) to define economic commensalism formally and compare its productive and ecological implications with predatory, extractive, and mutualistic resource-use regimes.

The analysis is guided by the following **general hypothesis**: When ecological functionality jointly determines productive outcomes and is itself affected by resource-use and preservation decisions, institutional arrangements that incorporate locally relevant ecosystem benefits and costs into private payoffs can admit a commensal configuration in which productive activity remains privately viable without eroding the ecological functions on which it depends.

This general hypothesis is decomposed into three **specific hypotheses**.

(H1) Productive-ecological endogeneity. When ecological functionality affects output and simultaneously responds to production, preservation, and institutional rules, treating it as a fixed input or exclusively as an external effect omits feedback mechanisms that alter marginal returns and may change the relative ranking of alternative resource-use strategies.

(H2) Local governance. When ecosystem functions are spatially heterogeneous and imperfectly substitutable, governance and compensation mechanisms anchored in ecologically relevant territorial units can reduce the divergence between private and territorial returns to preservation more effectively than spatially fungible mechanisms, provided

that relevant ecological effects are observable, local institutions possess sufficient implementation capacity, and cross-boundary spillovers are coordinated at broader institutional levels.

(H3) Intertemporal incentive compatibility. When private payoffs incorporate the contribution of ecological functionality to current and expected future production, remunerated ecosystem services, and avoided losses from degradation, the range of opportunity costs for which preservation is privately optimal becomes larger than under a payoff based exclusively on current marketed output. Preservation becomes individually rational whenever the resulting marginal private benefits equal or exceed its opportunity and implementation costs.

These hypotheses are analytical rather than directly empirical. H1 is examined by comparing the marginal conditions obtained under exogenous and endogenous representations of the ecological state; H2 by comparing alternative spatial and institutional configurations; and H3 by deriving the threshold at which the private return to maintaining ecological functionality becomes at least as large as the return to degradation. The dynamic stability of this configuration under uncertainty, learning, and repeated decisions is left to the subsequent intertemporal model.

The article makes three principal contributions. First, it distinguishes the inclusion of natural resources as conventional inputs from the endogeneity of ecological functionality as a jointly determined state of production. Second, it connects this productive representation to private payoffs and to governance arrangements anchored in ecologically relevant territorial units. Third, it defines economic commensalism through the joint behavior of private viability and ecological functionality and derives the conditions under which this configuration can emerge relative to predatory, extractive, and mutualistic regimes. Together, these contributions provide a conceptual and analytical basis for treating preservation not only as an external constraint, but as a potentially incentive-compatible component of productive strategies.

Methodologically, the article adopts a theoretical-analytical approach that combines production theory, environmental economics, and institutional analysis. It first reconstructs how externalities, property rights, common-pool resources, and natural capital illuminate distinct parts of the production-preservation problem. It then develops a preliminary model in which the environmental state enters the production structure and the payoff of productive agents, while incentives and governance affect the relative return to preservation and exploitation. The model is used to compare alternative ecological-economic strategies and to derive the conditions under which a commensal configuration can emerge as an individually rational and ecologically non-degrading outcome. The anal-

ysis remains deliberately foundational: it establishes the conceptual and static analytical basis upon which intertemporal dynamics, uncertainty, learning, and empirical testing can subsequently be developed.

The remainder of the article is organized as follows. Section 2 contextualizes the production-preservation problem and the limitations of current environmental governance. Section 3 reviews the relevant literature, moving from externalities and property rights to common-pool governance and natural capital, and identifies the elements that remain analytically disconnected. Section 4 develops the conceptual and analytical model of economic commensalism, defines the alternative ecological-economic regimes, and states the relevant equilibrium conditions. Section 5 presents the theoretical results and comparative implications of the model. Section 6 discusses their relationship to the existing literature, their institutional meaning, and the limitations of the proposed framework. Section 7 concludes and indicates directions for dynamic formalization and empirical assessment.

2. Problem Context

—> Mecanismos atuais de governança ambiental (regulação, precificação, mercados de carbono, etc.) são insuficientes para lidar com a complexidade e a dinâmica dos sistemas socioecológicos.

—> Dados e evidências mostram que, apesar dos esforços, as tendências de degradação ambiental e perda de biodiversidade continuam, indicando falhas estruturais nos modelos de governança atuais.

—> Citar esforços internacionais (Acordo de Paris, Objetivos de Desenvolvimento Sustentável) e nacionais (políticas ambientais, leis de proteção) que não têm conseguido reverter as tendências negativas.

—> Usar dados de desmatamento, perda de biodiversidade, emissões de gases de efeito estufa, Brasil e mundo, para ilustrar a insuficiência dos mecanismos atuais.

–> Concluir que é necessário repensar os fundamentos teóricos e práticos da governança ambiental, buscando abordagens que integrem melhor a natureza como parte dos sistemas econômicos e sociais, e que sejam mais adaptativas, inclusivas e eficazes.

3. Literature Review

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Posicionar explicitamente em relação a Dasgupta (2021): O relatório The Economics of Biodiversity (Dasgupta Review) endogeneiza a natureza de forma sofisticada. Mostrar no que seu framework vai além desse trabalho já canônico.

4. Conceptual and Analytical Framework

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; and (iv) to distinguish commensal resource use from predatory, extractive, and mutualistic configurations according to their joint effects on private viability and ecological functionality.

5. Theoretical Results

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6. Discussion

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7. Final Remarks

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